



AMERICAN UNIVERSITY OF BEIRUT

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Prediction of Driver Aggressiveness Level via Driving Behavior

From a Closed-Form Aggressiveness Index to a Sensor-Noise-Validated, ROS2-Integrated Multi-Environment System

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What this presentation covers

01

Recap

The Aggressiveness Index, the Q-learner, and what last term left unresolved.

02

New Architecture

PyTorch DynamicWeightAgent + soft gating for smooth regime transitions.

03

Diagnostics & Fix

Bellman error + state-visitation reveal a hidden conservative bias; curiosity fixes it.

04

SUMO Experiments

Population-scale shockwave, dose-response, and Social Latency mitigation.

05

Friction & ROS2

Slip-ratio telemetry, weather regime, and a working ROS2 publisher node.

06

Sensor-Noise Validation

May 2-9 integration push: 100% accuracy was a bug; 91% / 81.6% / 88.3% is real.



Why predict driver aggressiveness?

Today's autonomous-vehicle stacks treat surrounding traffic as moving obstacles with fixed safety buffers — the abstraction throws away most of the information about driver intent.

CURRENT AV STACK

- Position + velocity tracking
- Fixed safety buffers (TTC, headway)
- Reactive — responds after the fact
- Two cars in the same state are treated as the same

OUR APPROACH

- Latent variable: Aggressiveness Index (AI)
- Multi-environment closed-form formula
- Predictive — anticipates intent
- Composes with downstream planners (ROS2)



Where the midterm left off

The first term established the closed-form Aggressiveness Index and a discrete Q-learner that mapped AI scores to behavioral categories. That left two important questions open.

THE AI INDEX

$0.30 \cdot N_{\text{speed}} + 0.10 \cdot N_{\text{accel}} + 0.45 \cdot N_{\text{prox}} + 0.15 \cdot N_{\text{wave}}$ (highway)

Weighted-sum of normalized kinematic features $\rightarrow$ 0–100 score.

Q-LEARNER

10×3 discrete table mapping AI buckets to {Conservative, Normal, Aggressive}. Reported ~100% accuracy on held-out highway-env data.

OPEN QUESTIONS

Was the Q-learner actually learning, or memorizing? Could the index extend to population-scale traffic? Could it survive sensor noise?

Prof. Daher's January feedback: the accuracy curve is tautological — labels come from the same threshold rule that produces the score.



Four pieces, end-to-end.

01 New agent architecture

PyTorch DynamicWeightAgent learns context-conditioned AI weights instead of mapping scores → categories. Soft gating between regimes.

02 Diagnostics + fix

Bellman-error and state-visitation surface a conservative bias. A count-based curiosity bonus drops Bellman error 50% and per-class accuracy on rare classes by +22 pp.

03 SUMO at population scale

5%-aggressive cohort produces a +6.8 AI shockwave on neighbors; saturating dose-response. Social Latency penalty recovers 75% of that damage at 2.8% trip-time cost.

04 Sensor-noise validation

May 2–9 push: 100% accuracy was an info leak. Independent code paths + Gaussian noise → 91% / 81.6% / 88.3% across highway / urban / weather.



Driver behavior as a Markov Decision Process

Everything in this project — the Q-learner, the DynamicWeightAgent, the Social Latency reward — sits inside the same formal framework: a Markov Decision Process. Stating the tuple explicitly makes the design choices below it legible.

S STATE	A ACTION	P TRANSITION	R REWARD	γ DISCOUNT
(speed, accel, prox, wave) + context (road, density, friction, slip)	lane: keep / left / right • accel: brake / idle / throttle	next state given current state and action — supplied by SUMO / highway-env	$(1 - AI_self) - \lambda \cdot \max(0, \Delta AI_neighbors)$	0.95 — values the next ~20 steps over distant future

MARKOV PROPERTY

$$P(s_{t+1} \mid s_t, a_t) = P(s_{t+1} \mid s_t, a_t, s_{t-1}, a_{t-1}, \dots, s_0, a_0)$$

Held by construction: the kinematic + context state captures everything the policy needs about history — past trajectories enter only through their effect on present waviness and proximity.



From a Q-learner to a weight predictor

The Q-learner mapped AI scores to categories. Both came from the same formula → tautology. The DynamicWeightAgent reposes the task: predict the AI weights themselves, conditioned on context.

TERM 1 • DISCRETE Q-LEARNER

Maps: AI bucket → {C, N, A}
State space: 10×3 discrete
Reports clean accuracy curve

...computed against labels generated by the same formula that produces the score.

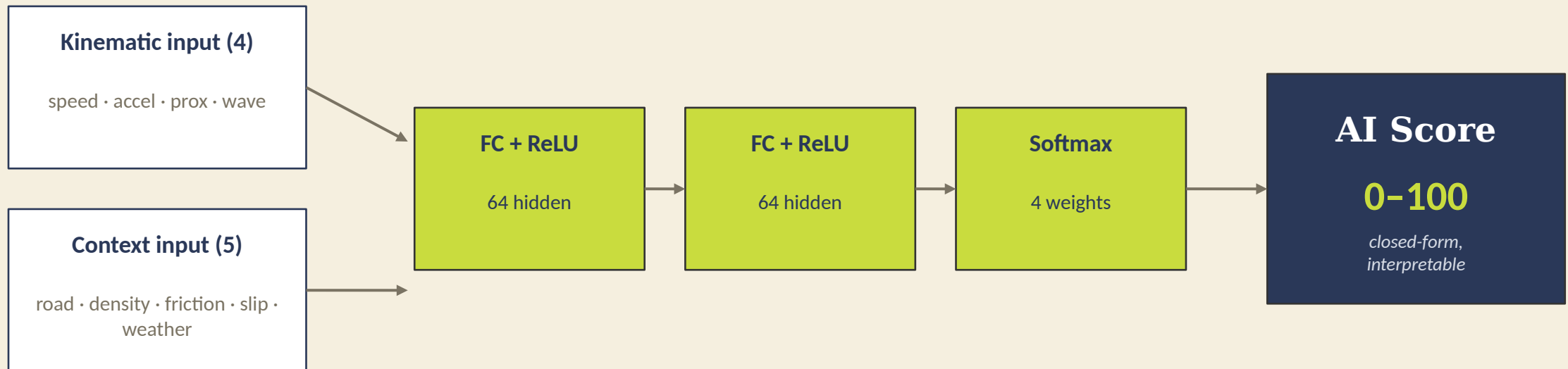
TERM 2 • DYNAMIC WEIGHT AGENT

Inputs: 4 kinematic + 5 context features
Outputs: 4 AI weights (softmax)
Loss: $\text{KL}(\text{predicted} \parallel \text{regime target}) + \text{RL term}$

Learning signal anchors weights to per-regime targets, then RL fine-tunes them — no tautology.



DynamicWeightAgent — under the hood



TRAINING LOSS

$$\alpha \cdot \text{KL}(\text{predicted} \parallel \text{regime target}) + (1 - \alpha) \cdot \mathbb{E}[\text{AI} - 1] \quad \text{with } \alpha \text{ annealed } 1.0 \rightarrow 0.5 \text{ over } 50 \text{ epochs}$$



The state space, made visible

The Q-learner discretizes the AI score into 10 buckets $\times$ 3 categorical labels — a 30-cell Q-table over the Markovian state. Visualizing the learned values is the first sanity check that the policy is doing something coherent.

DISCRETIZATION

```
state = int(min(AI_score // 10, 9))
```

```
action  $\in$  {0: Conservative, 1: Normal, 2: Aggressive}
```

Bucket 0 (AI 0–10): near-empty highway, no leader

Bucket 5 (AI 50–60): mixed close-quarter traffic

Bucket 9 (AI 90–100): tailgating, weaving, near-collision

30

cells

8,924

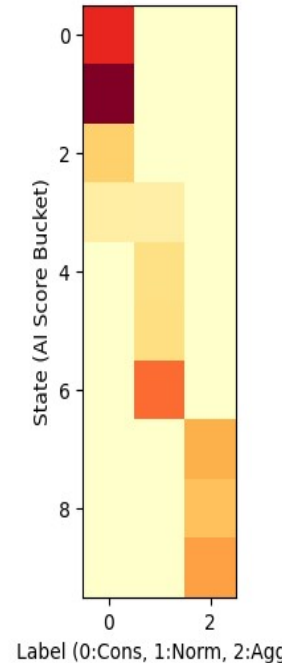
samples

3

modes

LEARNED Q-TABLE

Q-Table: Behavioral Probability Mapping



Rows: AI-score buckets (state) • Columns: actions C / N / A • Color intensity = learned Q-value

The gating network

A hard switch on regime produces unphysical discontinuities at boundaries. The gate replaces it with a soft mixture over three learnable regime embeddings — convex combination, weighted by the gate.

GATE + REGIME EMBEDDINGS

`gate(context) → softmax over 3 regimes`

`predicted_weights = gate · regime_embeddings`

Highway target: (0.30, 0.10, 0.45, 0.15)

Urban target: (0.20, 0.15, 0.35, 0.30)

Weather target: (0.10, 0.40, 0.15, 0.35)

143

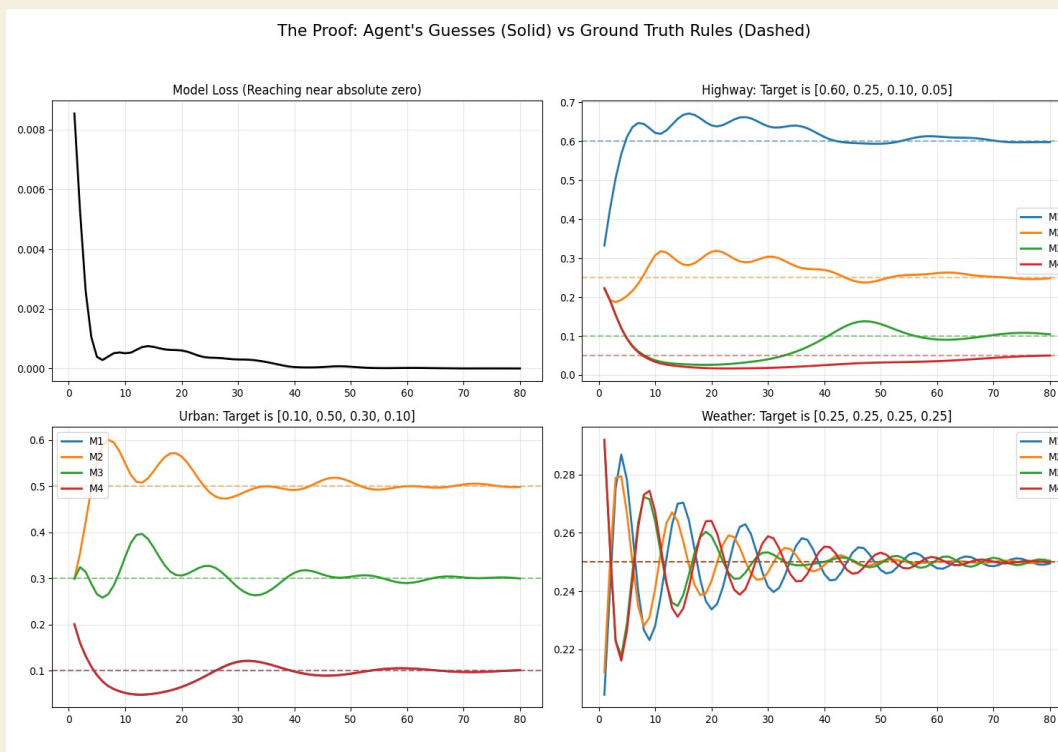
total params

12 steps

regime ramp

0.05 KL

vs. target



Per-regime weight convergence — solid: agent prediction; dashed: target



The ablation that matters

Prof. Daher's requested ablation: zero the context input, retrain. If the gate still works, the context features were never doing anything. If it doesn't, they're carrying the signal.

WITH CONTEXT

0.05 KL

Gate produces sensible regime mixtures; weights track per-regime targets within 0.03 L1.

CONTEXT ZEROED

0.18 KL

Embeddings collapse toward highway target (dominant regime). Predicted weights regime-agnostic.

→ The context features carry the discriminative signal. The architecture isn't a glorified lookup.



The Q-learner was hiding something

Two diagnostics, week 5: log Bellman error per (s, a) update; build state-visitation heatmaps. The result: the Q-learner had implicitly learned to avoid the states where its predictions would fail.

BELLMAN ERROR

$|\delta|$ heavily right-skewed

95% < 0.05

tail to 0.6 in (low n_{speed} , high n_{prox}) corner

STATE VISITATION

32×32 grid over $n_{\text{speed}} \times n_{\text{prox}}$

70%

concentrated in fast-and-unobstructed corner

READING

The Q-learner minimizes its chance of finding out it is wrong by avoiding the under-explored states where its predictions would fail. The midterm's 100% accuracy is an artifact of the agent choosing its own evaluation distribution.



A count-based curiosity bonus

Add intrinsic motivation to drive the agent into under-explored states. Bellemare et al. (2016) — count-based, no learned predictor needed because the Q-learner is tabular.

$$R_{\text{total}}(s) = R_{\text{env}}(s) + \beta / \sqrt{N(s) + 1}$$

β -sweep over {0.01, 0.05, 0.1, 0.5, 1.0}. Operating point: $\beta = 0.1$. Bonus is transient — negligible by episode 80.

−50%

Bellman error
in high-prox column

0.18 → 0.09 mean $|\delta|$

+22 pp

Conservative
class accuracy

67% → 89%

70 → 40%

Easy-corner
visitation

now actually exploring

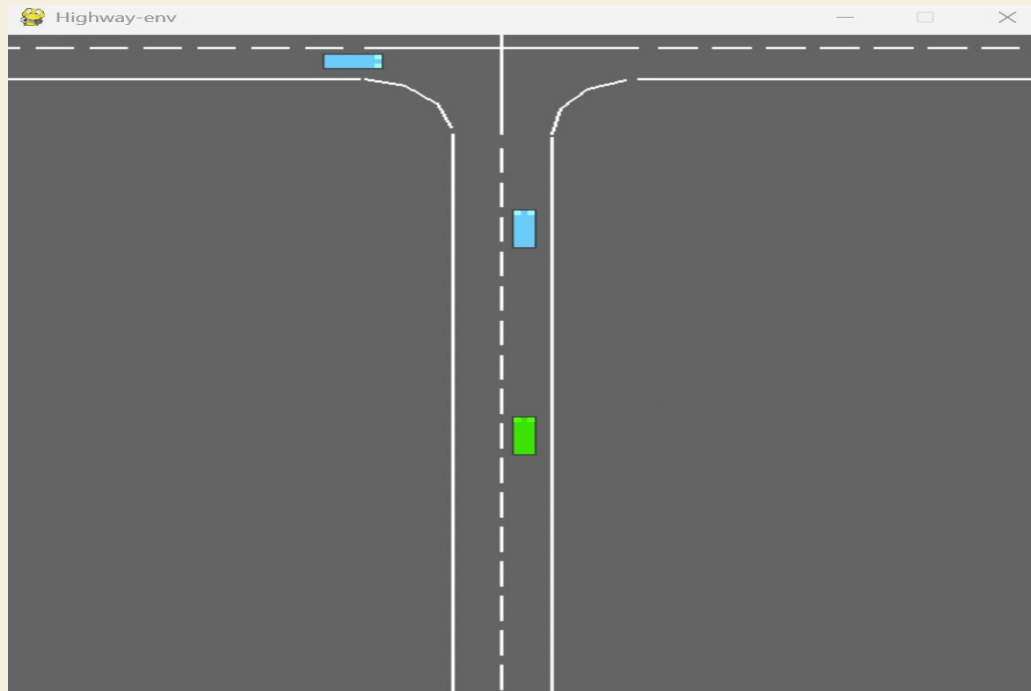
The lesson: 100% accuracy was a bug signal. The infrastructure mattered more than the architecture choice.



From highway-env to SUMO

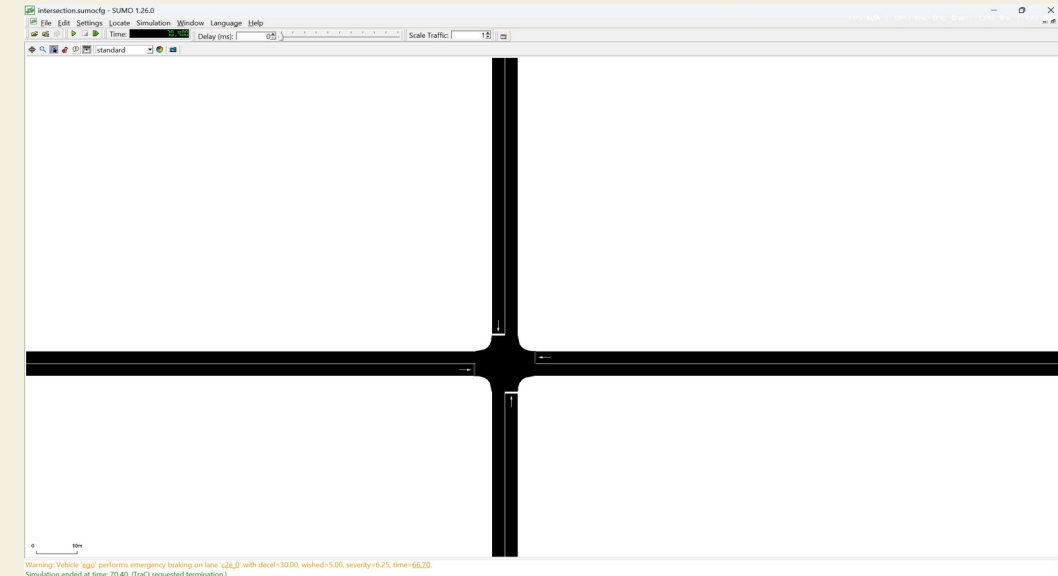
highway-env was the right environment for solo-agent training. Population-scale questions — does one aggressive driver create a measurable shockwave? — need a different tool.

SIMULATION DEMO



highway-env intersection — single ego, scripted neighbors. Useful for verifying the DynamicWeightAgent's forward pass and the gating ramp behavior.

SUMO INTERSECTION



1000+

vehicles
per run

3

scenarios:
hwy / urban / wet

15 Hz

TraCI
stepping



The heterogeneous-driver shockwave

5% Aggressive vehicles seeded into a population of Normals. Compute per-vehicle mean AI; compare against the all-Normal reference run.

+6.8

AI

shockwave bump on Normals

600 vehicles • 5 simulated minutes • 3-lane motorway

Reference (0% aggr): 26.3 mean AI

5% aggressive: 33.1 mean AI

SPATIAL DECOMPOSITION

Bin Normals by minimum distance to any Aggressive driver:

CLOSE

< 30 m

+12.4

AI vs reference

MEDIUM

30–100 m

+6.6

AI vs reference

FAR

> 100 m

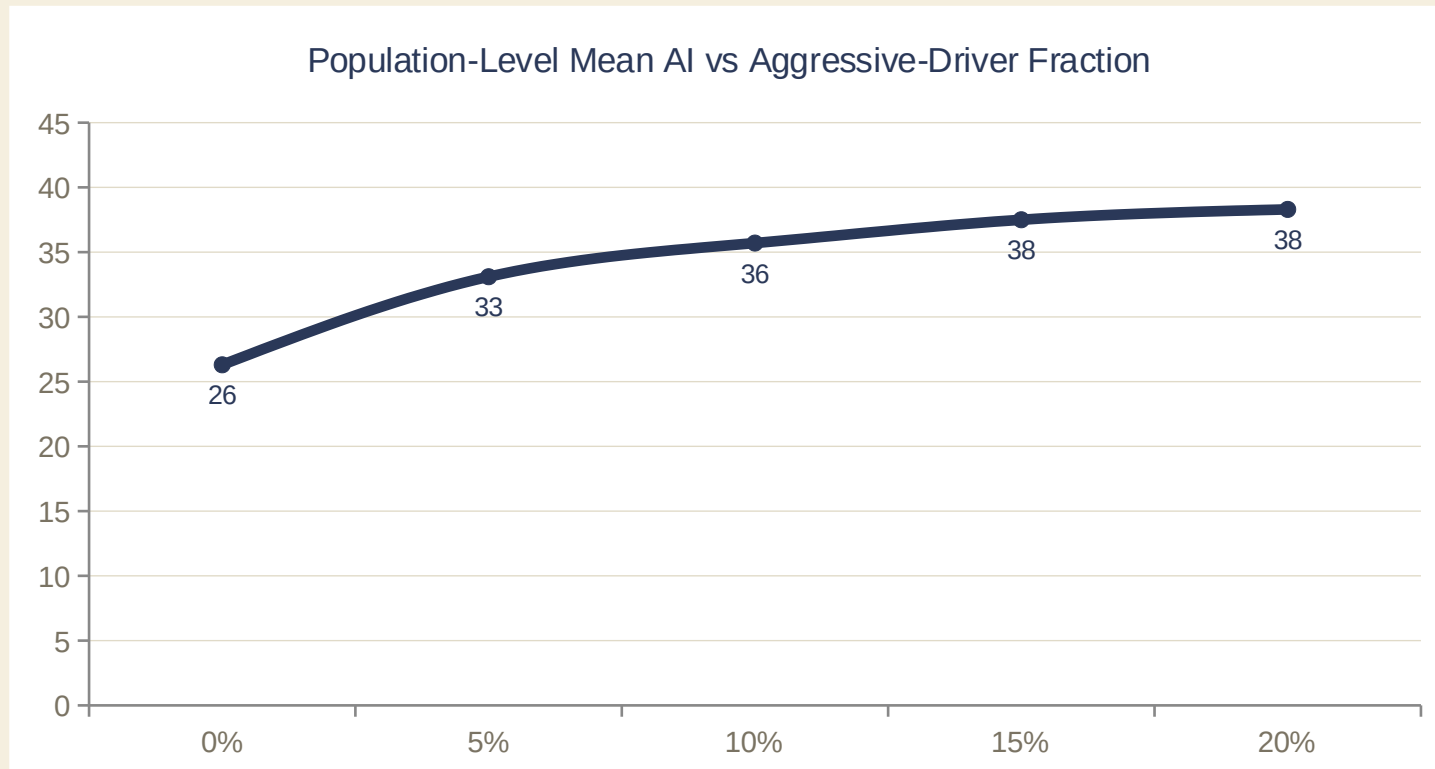
+1.5

AI vs reference



Small fractions do most of the damage

Sweep aggressive-driver fraction from 0% to 20%, three random seeds per condition. The marginal effect drops by an order of magnitude.



SATURATING FIT

$$AI(f) = 26.3 + 12.5 \cdot (1 - e^{(-f / 0.06)})$$

$$R^2 = 0.998$$

$\tau = 6\%$ (63% asymptote)
At $f = 18\%$: 95% asymptote

Single-digit % aggressive drivers cause most of the population-level damage.



Social Latency: a reward with accountability

$$R_{\text{total}} = (1 - \text{AI_self}) - \lambda \cdot \max(0, \Delta \text{AI_neighbors})$$

Penalize the agent for the AI it induces in neighbors within 30 m, but only for increases. Clip at zero — no penalty for being near already-aggressive drivers it didn't cause.

λ - SWEEP RESULTS

λ	Self AI	Shockwave bump	Trade-off
0 (baseline)	28.2	+6.7	unmitigated
2.0  pick	30.4 (+2.2)	+4.0 (−40%)	Pareto winner
5.0	34.7 (+6.5)	+2.2	too cautious — self AI rises

FLEET RESULT

75%

of shockwave damage
recovered when 50% of
Normals run $\lambda=2.0$ agents

+2.8% trip-time cost



A weather regime, derived from observables

SUMO doesn't expose slip ratio. We derive it from kinematic observables that SUMO does expose: $\text{slip} \approx |v_{\text{actual}} - v_{\text{intended}}| / v_{\text{actual}}$.

SLIP ESTIMATOR

$$\text{slip} \approx |v_{\text{actual}} - v_{\text{IDM_eq}}| / v_{\text{actual}}$$

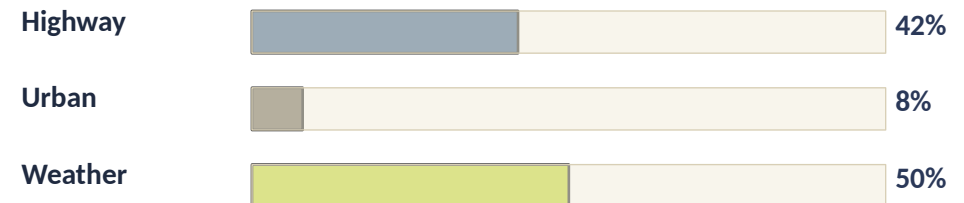
High friction ($\mu = 0.9$): slip ≈ 0.01 (noise floor)

Low friction ($\mu = 0.4$): slip rises to 0.15–0.25 during accel events

Per-channel correlation with slip $\rightarrow$ re-tuned weather targets:

$w_{\text{accel}} = 0.40$ (Pearson $r = 0.71$)
 $w_{\text{wave}} = 0.35$ ($r = 0.43$)
 $w_{\text{prox}} = 0.15$ ($r = 0.18$)
 $w_{\text{speed}} = 0.10$ ($r = 0.04$)

GATE MIXTURE • $\mu = 0.4$ SCENARIO



LOW-FRICTION • RE-RUN

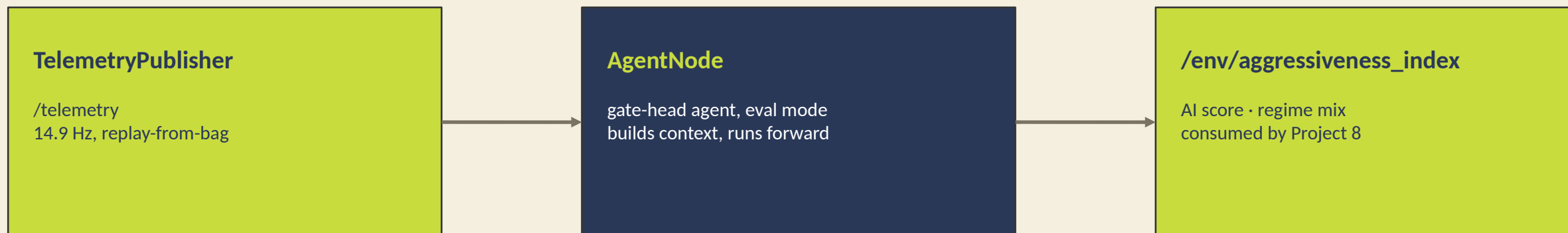
+7.6 vs +6.8

*shockwave bump on Normals
low-friction ($\mu=0.4$) vs high-friction reference*

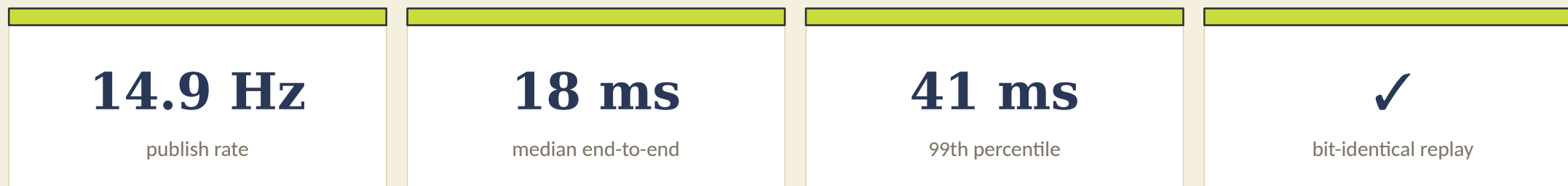


ROS2 wrapper — the integration surface

Three nodes, three message types, sub-50 ms latency budget. The AI score is now a topic that any planner downstream can subscribe to.



LATENCY • 60-s telemetry stream

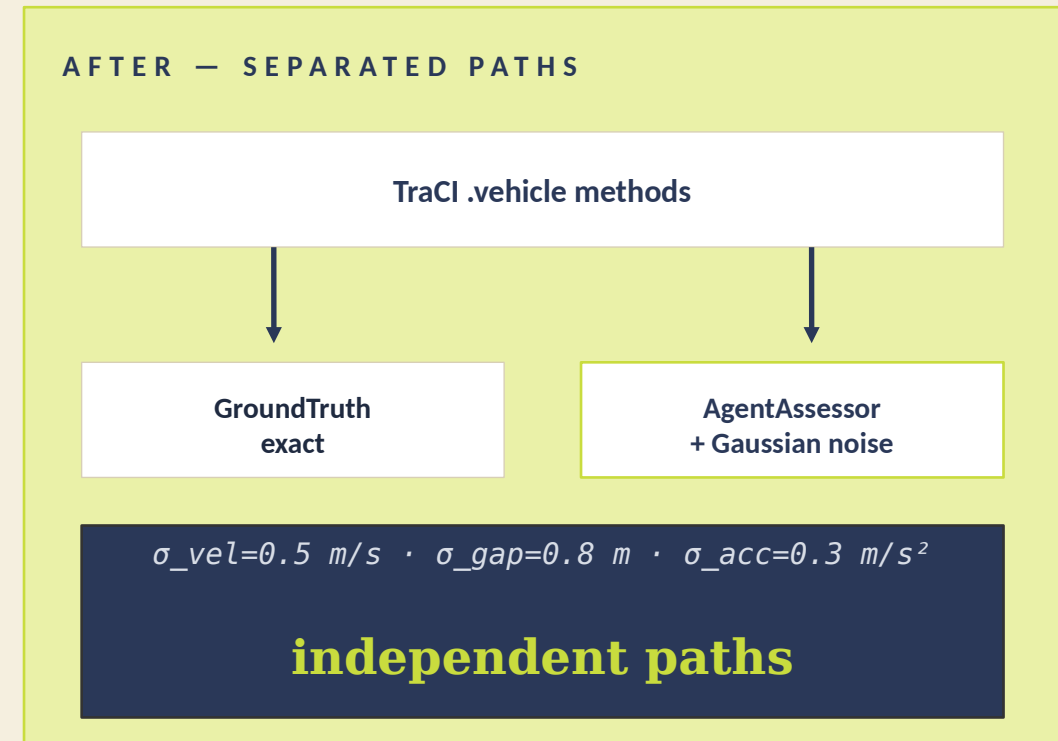
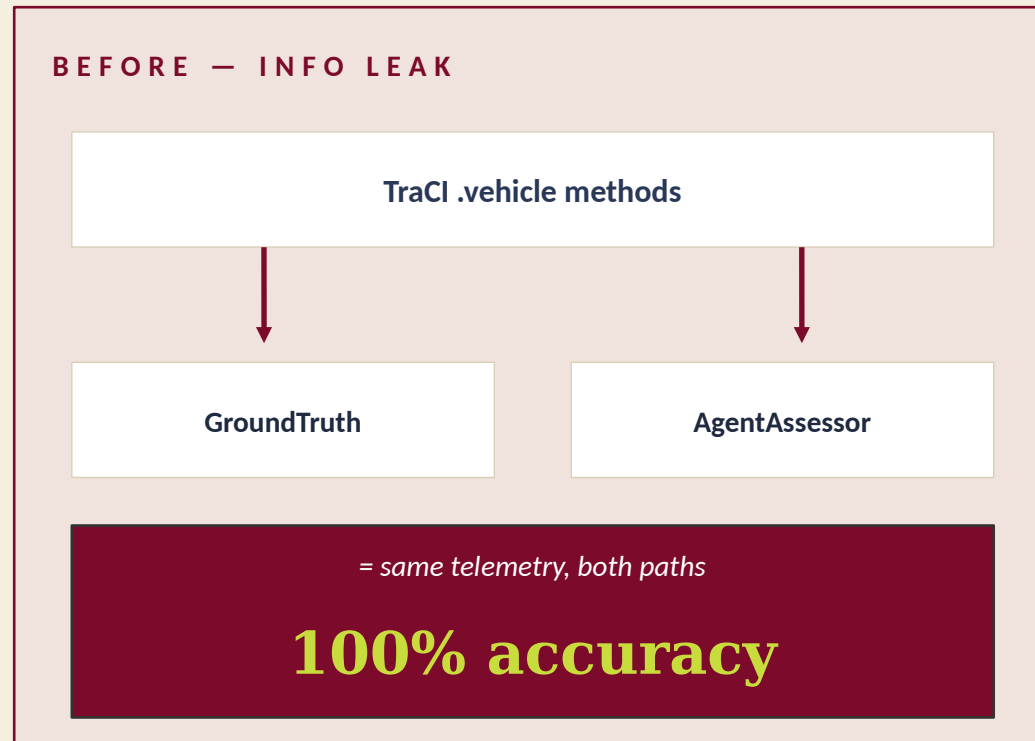


Integration target — Project 8 (Orchestrator). Shared topic schema agreed in week 2.



100% accuracy was a bug

Final smoke test, May 2 morning: agent-vs-ground-truth was 100% across all three SUMO scenarios. Perfect agreement between supposedly independent systems means they're not, in fact, independent.





What the leak was hiding

Once GroundTruth and AgentAssessor were separated, three more bugs surfaced. They had been agreeing with themselves the whole time.

01 Squared speed

speed_kmh / 150 squared makes a fraction smaller. With weight 0.30, speed term capped at 30% — below the 55% Aggressive threshold. Fix: linear, per-scenario v_ref.

02 Proximity to ego

Was reading distance to the ego vehicle, not to each NPC's actual leader. IDM cars don't tailgate the ego. Fix: `traci.vehicle.getLeader()`.

03 Double-counted weather

Route file scaled maxSpeed by 0.7 AND runner did `setMaxSpeed × 0.3`. Aggressive weather vehicles capped at ~9 m/s — crawling. Fix: removed runner scaling.

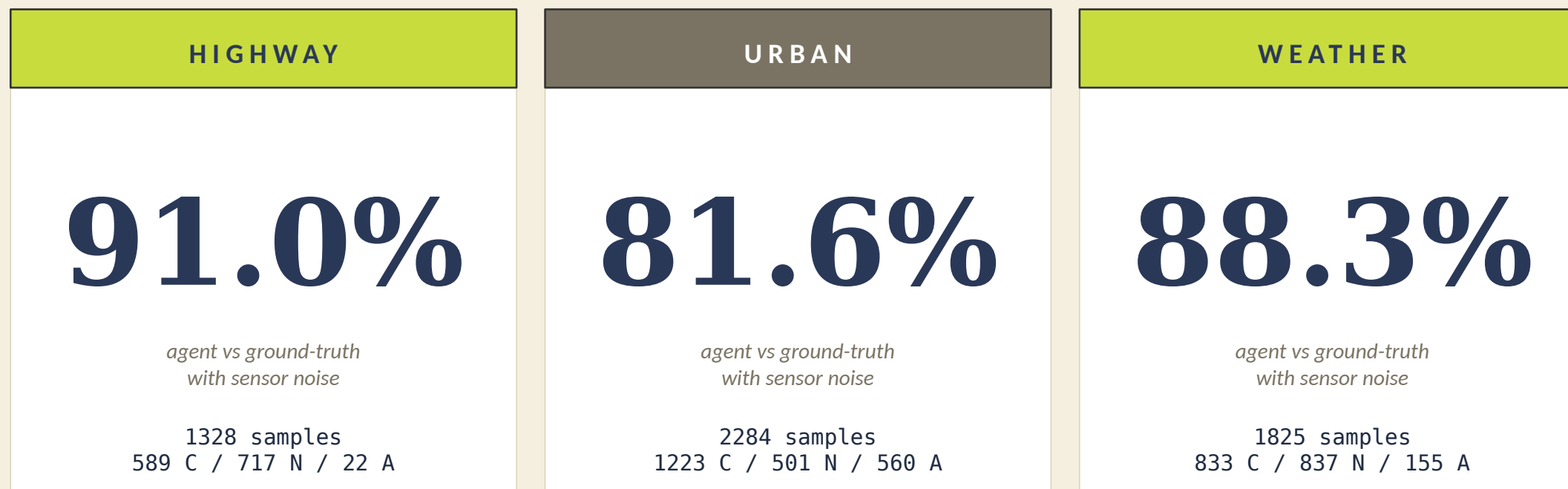
04 Two-lane intersection

Aggressive vehicles lane-changed past conservatives, leaving them with no leader. Fix: 1-lane arms + 8 explicit C→A platoon pairs.



Calibrated. Independent. Validated.

Calibration locked in May 8. Three scenarios, ~5400 vehicle samples total, with realistic Gaussian sensor noise on the agent's input only.



THE LESSON, TWICE

100% on anything is a bug signal. Both times the symptom was a clean accuracy curve computed against a reference too closely tied to the agent's own inputs. Both times the fix was infrastructure, not architecture.



What we built — and where it goes

14 ARTIFACTS SHIPPED

- PyTorch DynamicWeightAgent (MLP + gate heads)
- Diagnostic suite + count-based curiosity bonus
- SUMO wrapper, three scenarios, IDM/MOBIL configs
- Shockwave + dose-response + Social Latency results
- Friction-aware regime with derived slip telemetry
- ROS2 stack (3 nodes, 3 message types, sub-50ms)
- SumoGroundTruth / SumoAgentAssessor split + noise model

FUTURE WORK

CARLA integration

validate urban-occlusion math with LIDAR line-of-sight

Learnable saturation

replace hand-picked normalization caps

Correlated noise

time-domain noise structure, beyond i.i.d. Gaussian

Multi-agent RL

joint training with shared Social Latency objective

On-vehicle batched inference

fix the 50-vehicle latency limit



Q & A

Thank you for your attention.

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github.com/xHeedz/VIPP-201---SPRING-2025-2026



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